

Scheme 6

BPS Ontology and Semantic Loading Benchmark Scheme

Ontology prompts for benchmark-relative semantic quantification

DRAFT FOR EXPERT EVALUATION. These coding schemes are a working draft circulated for expert evaluation. They have not been applied to a final review corpus. The current manuscript is a test run that exercised an earlier, coarser generation of these schemes in the Python workflow with an LLM (gemini-2.5-flash); the full end-to-end run is deliberately held until after expert feedback, and the refinements proposed here are awaiting sign-off before any re-run.

At a Glance

Workflow position

Semantic loading and synthesis after Stage 2 coding.

Operational mode

Ontology-prompted embedding benchmark, with OpenRouter embeddings when available and TF-IDF fallback otherwise.

Unit of analysis

One composed record string (title, abstract, objective) scored against domain and subdomain prompts.

Provenance basis

semantic_loading.py and the archived ontology terms JSON (openai/text-embedding-3-small).

Research questions

RQ2 (scope and balance via benchmark-relative domain loading)

Tagline

Ontology-prompted embedding benchmark with TF-IDF fallback

Canonical Source Paths

- `src/bps_review/reporting/semantic_loading.py`
- `src/vector_db/semantic_loading/ontology/ontology_terms.json`
- `src/vector_db/semantic_loading/analysis/domain_loading_summary.csv`
- `src/vector_db/semantic_loading/analysis/subdomain_loading_summary.csv`

Purpose

This scheme supplies the ontology scaffold used to quantify semantic emphasis across biological, psychological, and social axes. It is not a manual adjudication form, but it is an operational text-classification framework: it standardizes the domain and subdomain prompts against which review records are embedded and compared.

Because loadings are benchmark-relative, the ontology prompts are the measuring instrument. Their wording and coverage directly determine the domain-balance results, so they warrant expert scrutiny.

Scoring Construction

How a record is scored.

Record text

Title, abstract, and extracted objective text composed into one semantic analysis string.

Domain prompt

{domain} chronic pain ontology. {joined subdomains}.

Biological prompt assembly

Shared biological core plus the active track's extension: shared core plus musculoskeletal for the musculoskeletal review, shared core plus neuropathic for the neuropathic review. Psychological and social prompts are identical across both tracks.

Subdomain prompt

{domain} chronic pain subdomain. {term}.

Domain order

Always biological, psychological, social.

Embedding

OpenRouter openai/text-embedding-3-small when available; TF-IDF cosine fallback otherwise.

Biological Ontology: Shared Core Plus Track Extensions

The biological pole is the one place where a single uniform subdomain list would distort the science, because the biological mechanisms of musculoskeletal and neuropathic pain genuinely differ. The biological ontology is therefore built as a shared core that applies to both reviews, plus a musculoskeletal extension used only for the musculoskeletal review and a neuropathic extension used only for the neuropathic review.

When a record is scored, the biological domain prompt is assembled from the shared core plus the active track's extension. The psychological and social ontologies below are identical for both reviews. This keeps the two reviews comparable on the psychological and social axes while measuring each one's biology against the mechanisms that actually matter for it.

Biological Subdomains: Shared Core (9)

Applied to both the musculoskeletal and neuropathic reviews.

- Central Sensitization and Neuroplasticity
- Nociceptive and Pain Signalling Pathways
- Immune Inflammatory and Neuroinflammatory Processes
- Neuroimaging Brain Structure and Function
- Genetic Epigenetic and Biological Vulnerability
- Sleep Disruption and Circadian Dysregulation
- Pharmacological and Biomedical Treatment
- Metabolic Nutritional and Hormonal Factors
- Physical Function Mobility and Deconditioning

Biological Extension: Musculoskeletal (3)

Added only for the musculoskeletal review.

- Musculoskeletal and Structural Pathology
- Joint Degeneration and Osteoarthritic Change
- Muscle Tendon and Soft Tissue Pathology

Biological Extension: Neuropathic (5)

Added only for the neuropathic review.

- Peripheral Nerve Injury and Neuropathy
- Sensory Phenotype and Quantitative Sensory Testing
- Ectopic Firing Ion Channels and Neuronal Excitability
- Small Fiber and Nerve Conduction Pathology
- Deafferentation and Central Neuropathic Mechanisms

Psychological Subdomains (20, uniform across both reviews)

Identical for the musculoskeletal and neuropathic reviews.

- Catastrophizing and Negative Cognitive Appraisal
- Fear Avoidance and Pain Related Fear
- Depression Emotional Distress and Affect
- Anxiety and Psychological Reactivity
- Self Efficacy Control Beliefs and Perceived Mastery
- Acceptance Psychological Flexibility and Mindfulness
- Pain Coping Strategies and Adjustment
- Attention Vigilance and Pain Processing
- Illness Beliefs Pain Representations and Meaning
- Cognitive Behavioral and Psychotherapeutic Approaches
- Third Wave Therapies ACT and Contextual Approaches
- Resilience Positive Psychology and Post Traumatic Growth
- Identity Self Concept and Chronic Pain Biography
- Trauma Adverse Childhood and Life Events
- Personality Psychological Traits and Individual Differences
- Cognitive Function Executive Processes and Brain Health
- Motivational Processes Goal Pursuit and Engagement
- Healthcare Seeking Treatment Adherence and Engagement

- Emotional Regulation and Pain Affect Processing
- Mental Health Comorbidity and Psychological Wellbeing

Social Subdomains (12, uniform across both reviews)

Identical for the musculoskeletal and neuropathic reviews.

- Social Support Network and Interpersonal Resources
- Work Disability Occupational Function and Productivity
- Family Caregiver and Household Dynamics
- Socioeconomic Status and Health Inequity
- Healthcare Access Navigation and System Factors
- Cultural Ethnic and Demographic Context
- Community Participation and Social Role Functioning
- Legal Compensation and Medicolegal Systems
- Health Literacy Education and Patient Empowerment
- Stigma Social Isolation and Exclusion
- Return to Work Vocational Rehabilitation and Employment
- Social Determinants of Pain and Environment

Proposed Refinement: Seed-Term Expansion and Coverage Audit

PROPOSED REFINEMENT

Awaiting expert evaluation. Not yet applied to the pipeline.

Refinement proposed for expert evaluation. Each subdomain prompt is currently just the subdomain label, so the benchmark is only as rich as those short phrases. We propose attaching three to five expert-curated seed terms to every subdomain (for example Central Sensitization and Neuroplasticity would carry windup, temporal summation, descending modulation, cortical reorganization). Richer prompts sharpen the semantic contrast between neighbouring subdomains and reduce the chance that a record loads on a subdomain by generic vocabulary alone.

We propose a short operational definition per subdomain so the boundaries are documented and reviewable, and a coverage audit that checks the three domain prompts for comparable length and specificity. An asymmetry in prompt richness across domains can by itself bias benchmark-relative loadings, so the audit protects the central social-shortfall finding from being an artefact of the instrument. The audit must also check that the musculoskeletal and neuropathic biological prompts (shared core plus each extension) are comparably rich, so neither review’s biology is under-measured relative to the other.

Finally we propose publishing the scoring transparency: the cosine to each domain prompt, the normalization applied, and the benchmark-relative redistribution step, so the loadings are fully reproducible from the prompts.

Interpretive Note

This is an analytic coding scheme, not a human extraction codebook. It is included because the project uses ontology prompts as a standardized semantic coding scaffold, and because the instrument’s design determines the domain-balance results that answer RQ2.

Primary Outputs Using This Scheme

- Record-level domain and subdomain loadings
- Domain summary and subdomain summary tables
- Pairwise loading analyses
- Dominance-by-review-type summaries